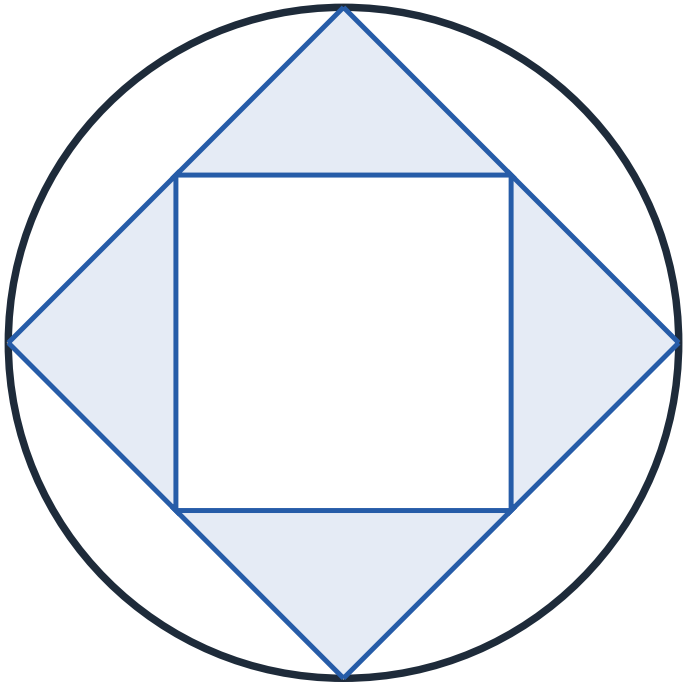
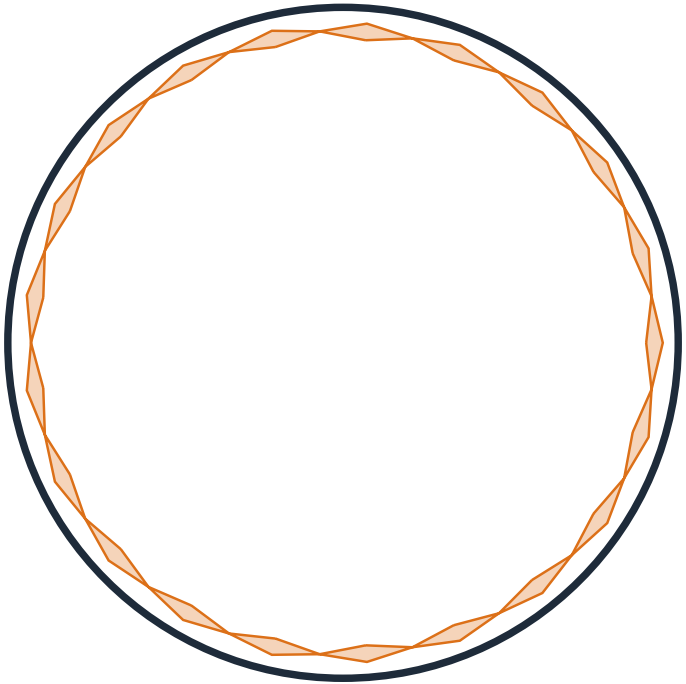


$N = 4 : \quad S_1 = r^2$



$N = 21, m = 15 : \quad S_2 \simeq 0.14463434 r^2$